Vijaya Lakshmi Bollapalli Data Scientist

Summary

Aspiring Data Scientist and AI/ML enthusiast with strong foundations in Python, Machine Learning. Experienced in data preprocessing, model building, and evaluation using Scikit-learn through academic projects. Skilled in SQL, data visualization, and extracting insights from structured datasets. Passionate about applying AI-driven solutions to solve real-world problems.

Education

Master of Science - Data Science 2023 – 2025 | CGPA: 7.20
Bankatlal Badruka College for Information and Technology

Bachelor of Science - Computer Science 2020 – 2023 | CGPA: 9.25
Andhra Mahila Sabha Arts and Science College for Women

Skills

Python & Data Analysis: NumPy, Pandas, Data Cleaning, EDA, Matplotlib, Seaborn

Machine Learning: Supervised Learning, Unsupervised Learning, Ensemble Learning, Model Evaluation & Validation, Hyperparameter Tuning, Cross Validation

Deep Learning: ANN, RNN, LSTM, GRU, NLP

Gen AI: LLM, Open AI, Gemini, Lang Chain, Prompt Engineering, RAG

Projects

Uber Ride Booking Status Prediction
- Developed an Uber Ride Booking Status Prediction system using Python and Artificial Neural Networks (ANN) on 150,000+ ride booking records.
- Performed data cleaning, feature engineering, missing value handling, encoding, and feature scaling to improve model performance.
- Applied SMOTE for class balancing and optimized the ANN model using Optuna hyperparameter tuning, achieving 91% accuracy.
- Built and deployed a Streamlit web application to predict booking outcomes such as Completed, Cancelled by Customer, Cancelled by Driver, Incomplete, and No Driver Found.

Fake News Detection Using NLP & Deep Learning
- Developed an end-to-end Fake News Detection System using Python, NLP, Word2Vec, and Bi-LSTM, classifying over 40,000+ news articles as real or fake.
- Performed text preprocessing and feature engineering, including tokenization, stop-word removal, stemming, and text vectorization, reducing data noise by 90%+.
- Built and optimized a Bidirectional LSTM (Bi-LSTM) model with Word2Vec embeddings, achieving 95%+ classification accuracy on unseen data.
- Evaluated model performance using Accuracy, Precision, Recall, and F1-Score, and deployed the trained model for real-time fake news prediction.

Zomato AI Food Assistant Chatbot
- Built an AI-powered food recommendation chatbot using Python, LangChain, GPT-4o-mini, and Streamlit.
- Developed context-aware conversations using prompt templates and memory management for personalized user experiences.
- Enabled restaurant recommendations, cuisine discovery, dietary preference support, and food-related query resolution through LLM-based natural language understanding.
- Deployed a real-time conversational application integrating OpenAI APIs and LangChain orchestration components.

AI Tutor – Learning Assistant Chatbot
- Developed an AI-powered learning assistant using Python, LangChain, OpenAI GPT-4o-mini, and Streamlit to answer technical questions in real time.
- Built a system to automatically classify questions into topics such as Python, SQL, Machine Learning, Power BI, Excel, and Data Analysis.
- Implemented keyword extraction and generated easy-to-understand explanations, sample code, and learning resources using Large Language Models (LLMs).
- Used prompt engineering and JSON output parsing to provide structured and accurate responses for users.
- Designed and deployed an interactive Streamlit web application to support personalized learning and mentor recommendations.

Certificates

Python, SQL, EDA and Power BI
Innomatics Research Labs

Data Visualization with Python
IBM (Cognitive Class)

Microsoft Power BI
Computer Society of India (CSI)